

DEPLOYMENT.md

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TL;DR

```
# Clone and install
git clone <your-repo-url> athena && cd athena

npm install
```

Build and run

```
npm run build
```

```
npm start -- --help
```

Prerequisites

Tool	Version	Install Link
-----	-----	-----
Node.js	≥18.0.0	https://nodejs.org
npm	≥9.0.0	(comes with Node.js)
Git	Any recent	https://git-scm.com
Anthropic API Key	N/A	https://console.anthropic.com

Optional (for diagram generation):

- Mermaid CLI: `npm install -g @mermaid-js/mermaid-cli`

Optional (for screenshots):

- Playwright browsers: Installed automatically during `npm install`

Step-by-Step Setup

1. Clone the Repository

```
git clone <your-repo-url> athena
cd athena
```

2. Install Dependencies

```
npm install
```

This will install:

- `@anthropic-ai/sdk` for Claude API integration
- `commander` for CLI parsing
- `diff` for change detection
- `glob` for file scanning
- `playwright` for screenshot generation

3. Build the Project

```
npm run build
```

This compiles TypeScript files from `src/` to `dist/`.

4. Set Up Environment Variables

Create a `.env` file in the project root:

```
cat > .env << 'EOF'
ANTHROPIC_API_KEY=sk-ant-api03-...

EOF
```

Or export directly:

```
export ANTHROPIC_API_KEY="sk-ant-api03-..."
```

5. Verify Installation

```
npm start -- --version
npm start -- --help
```

You should see the Athena version and available commands.

6. Run Your First Scan

```
npm start -- scan /path/to/your/project
```

Or in development mode:

```
npm run dev -- scan /path/to/your/project
```

Environment Variables

Variable	Description	Example Value	Required?
-----	-----	-----	-----
ANTHROPIC_API_KEY	Claude API key for documentation generation	sk-ant-api03-xxxxx	✔ Yes (for generation)
NODE_ENV	Environment mode	production	✗ No (defaults to development)
DEBUG	Enable debug logging	athena:*	✗ No

Getting your API key:

1. Go to <https://console.anthropic.com>
2. Navigate to API Keys section
3. Create a new key or copy existing one
4. Has billing access for production use

Running in Production

Option 1: Direct Node Execution

```
# Build once  
npm run build
```

Run as a global command

```
npm link
```

Now use from anywhere

```
athena scan /path/to/project
```

```
athena generate --docs
```

Option 2: PM2 (for long-running services)

```
npm install -g pm2
```

Start as daemon (if running as a

```
pm2 start dist/cli.js --name athena -- scan /path/to/project --watch
```

Monitor

```
pm2 logs athena
```

```
pm2 status
```

Option 3: Systemd Service

Create `/etc/systemd/system/athena.service` :

```
[Unit]
Description=Athena Documentation Generator

After=network.target

[Service]

Type=simple

User=your-user

WorkingDirectory=/path/to/athena

Environment="ANTHROPIC_API_KEY=sk-ant-api03-xxxxx"

ExecStart=/usr/bin/node /path/to/athena/dist/cli.js scan /path/to/target --w

Restart=on-failure

RestartSec=10

[Install]
```

```
WantedBy=multi-user.target
```

Enable and start:

```
sudo systemctl daemon-reload
sudo systemctl enable athena

sudo systemctl start athena

sudo systemctl status athena
```

Option 4: Cron Job (scheduled scans)

```
# Edit crontab
crontab -e
```

Add line (runs daily at 2 AM)

```
0 2 * cd /path/to/athena && /usr/bin/node dist/cli.js scan /path/to/project
```

Docker

Dockerfile

```
FROM node:18-alpine
```

Install Playwright dependencies

```
RUN apk add --no-cache \

    chromium \
    nss \
    freetype \
    harfbuzz \
    ca-certificates \
```

```
ttf-freefont
```

```
ENV PLAYWRIGHT_SKIP_BROWSER_DOWNLOAD=1
```

```
ENV PLAYWRIGHT_CHROMIUM_EXECUTABLE_PATH=/usr/bin/chromium-browser
```

```
WORKDIR /app
```

Copy package files

```
COPY package*.json ./
```

Install dependencies

```
RUN npm ci --only=production
```

Copy source and build

```
COPY tsconfig.json ./
```

```
COPY src ./src
```

```
RUN npm install -g typescript && \
```

```
    npm run build && \  
    npm uninstall -g typescript
```

Cleanup

```
RUN rm -rf src tsconfig.json
```


Runtime environment

```
ENV NODE_ENV=production
```

```
ENTRYPOINT ["node", "dist/cli.js"]
```

```
CMD ["--help"]
```

Build and Run

```
# Build image
docker build -t athena:latest .
```

Run with API key

```
docker run --rm \

  -e ANTHROPIC_API_KEY="sk-ant-api03-xxxxx" \
  -v /path/to/target-project:/project \
  athena:latest scan /project
```

Run with .env file

```
docker run --rm \

  --env-file .env \
  -v /path/to/target-project:/project \
  athena:latest scan /project --output /project/docs
```

Docker Compose

Create `docker-compose.yml` :

```
version: '3.8'
```

```
services:
```

```
  athena:
    build: .
    image: athena:latest
    environment:
      - ANTHROPIC_API_KEY=${ANTHROPIC_API_KEY}
    volumes:
      - ./target-project:/project
      - ./output:/output
    command: scan /project --output /output
```

Run:

```
docker-compose up
```

Troubleshooting

Issue: `ANTHROPIC_API_KEY` not found

Symptom:

```
Error: ANTHROPIC_API_KEY environment variable is required
```

Fix:

```
export ANTHROPIC_API_KEY="sk-ant-api03-xxxxx"
```

Or add to .env file

Issue: TypeScript build fails

Symptom:

```
error TS2307: Cannot find module '@anthropic-ai/sdk'
```

Fix:

```
rm -rf node_modules package-lock.json
npm install
```

```
npm run build
```

Issue: Playwright browsers not installed

Symptom:

```
Error: browserType.launch: Executable doesn't exist at /path/to/chromium
```

Fix:

```
npx playwright install chromium
```

Or install all browsers

```
npx playwright install
```

Issue: Port already in use (for dev server)

Symptom:

```
Error: listen EADDRINUSE: address already in use :::3000
```

Fix:

```
# Find and kill process using port  
lsof -ti:3000 | xargs kill -9
```

Or use a different port (if conf.

```
PORT=3001 npm start
```

Issue: Permission denied on `npm link`

Symptom:

```
EACCES: permission denied, symlink
```

Fix:

```
# Option 1: Use sudo (not recommended)
sudo npm link
```

Option 2: Fix npm permissions

```
mkdir ~/.npm-global

npm config set prefix '~/.npm-global'

echo 'export PATH=~/.npm-global/bin:$PATH' >> ~/.bashrc

source ~/.bashrc

npm link
```

Issue: Mermaid diagrams not generating**Symptom:**

```
Warning: Mermaid CLI not available, skipping diagram generation
```

Fix:

```
# Install globally
npm install -g @mermaid-js/mermaid-cli
```

Verify installation

```
mmdc --version
```

Issue: Out of memory during build

Symptom:

```
FATAL ERROR: Reached heap limit Allocation failed - JavaScript heap out of m
```

Fix:

```
# Increase Node.js memory limit
NODE_OPTIONS="--max-old-space-size=4096" npm run build
```

Issue: Git operations fail

Symptom:

```
Error: fatal: not a git repository
```

Fix:

This happens when scanning a non-git project. Either:

```
# Initialize git in target project
cd /path/to/project

git init
```

Or skip git-dependent features (

```
athena scan /path/to/project --no-git
```

Debugging Tips

Enable verbose logging:

```
DEBUG=athena:* npm start -- scan /path/to/project
```

Check TypeScript compilation:

```
npx tsc --noEmit
```

Validate config file:

```
cat .athenarc.json | jq .
```

Should output valid JSON

Test API connectivity:

```
curl https://api.anthropic.com/v1/messages \
  -H "x-api-key: $ANTHROPIC_API_KEY" \
  -H "anthropic-version: 2023-06-01" \
  -H "content-type: application/json" \
  -d '{"model": "claude-3-5-sonnet-20241022", "max_tokens": 1024, "messages": [{
```

Related Documentation

- [Readme](#)
- [Architecture](#)

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